Background Reference

BLAST and Sequence Similarity

From biological intuition to command-line analysis

Computing for Data-Driven Biology · 5BI00A
Master's Programme in Bioinformatics · Umeå University

Session Aims & ILO Links

1	Explain what sequence similarity is and why it is biologically meaningful	ILO 5
2	Describe how BLAST works — seeding, extension, scoring, and E-values	ILO 1
3	Interpret a BLAST results table including E-values, bit scores, and % identity	ILO 7
4	Distinguish the five BLAST programmes and select the appropriate one	ILO 5
5	Choose an appropriate BLOSUM matrix for a search — default, distant, or close relatives	ILO 5
6	Recognise common interpretation pitfalls — coverage vs. identity, and database-size effects	ILO 7

This is background reading, not a taught session. Read it before the "Three ways to run BLAST" hands-on exercise in Lecture 15, where you will use your TP53 sequence from Lecture 11.

Part 1

Sequence similarity — the biological foundation

Similarity Implies Common Ancestry

Sequences that are similar are likely related by evolution — they are homologous.
Homologous sequences often share biological function because structure is conserved by natural selection.

🧬 Orthologues

Genes in different species descended from the same ancestral gene via speciation. Usually retain the same function. This is the basis of functional annotation transfer across species.

🔀 Paralogues

Genes within the same genome related by gene duplication. Often diverge in function — one copy may retain the ancestral role while the other acquires a new one. Be cautious about annotation transfer between paralogues.

⚠ **Similar sequence ≠ identical function.**
The top BLAST hit is a hypothesis about function, not a fact. The strength of that hypothesis depends on E-value, % identity, query coverage, and the reliability of the subject sequence's own annotation.

Part 2

How BLAST works — a conceptual overview

The Problem BLAST Solves

You have one query sequence. The database has billions. How do you find all similar sequences quickly?

Naïve approach

Compare every residue of the query to every residue of every sequence in the database. Guaranteed to find the optimal alignment — but computationally impossible for large databases.

BLAST — heuristic approach

Use short-word seeds to rapidly identify candidate matches, then only extend full alignments from those seeds. Fast enough for billions of sequences. Does not guarantee finding the optimal alignment — but finds the biologically important ones.

Heuristic = fast and good enough. Not exhaustive. Understanding this limitation is part of interpreting BLAST results responsibly.

Seed and Extend: The Two-Stage Strategy

1 Seeding

Break the query into short overlapping words (k-mers). Find exact or near-exact matches to these words in the pre-indexed database. Fast — uses a lookup table. Each match is a seed.

2 Extension

Extend each seed in both directions, one residue at a time, as long as the alignment score keeps improving. Stop when the score drops too far below the best seen. The result is a High-Scoring Pair (HSP).

3 Reporting

Rank all HSPs by score. Calculate E-values. Report those above the significance threshold. Multiple HSPs for the same subject sequence may be merged.

Word size k: typically 11 nt for blastn, 3 aa for blastp. Smaller k = more sensitive but slower. Larger k = faster but may miss distant homologues.

Scoring: Substitution Matrices

Protein BLAST scores are not simply +1 for match, −1 for mismatch. Each amino acid pair has its own score based on how often they are observed to substitute for one another in real, related proteins.

BLOSUM matrices

Derived from observed substitutions in conserved blocks of related proteins.

BLOSUM80 Closely related sequences (< 80% identity used)

BLOSUM62 Moderately diverged — DEFAULT for most searches

BLOSUM45 Distantly related sequences — more sensitive

What the number means

BLOSUM62 is derived from sequences sharing ≤62% identity. The lower the number, the more diverged the reference sequences — and the better the matrix is at detecting distant relationships.

Conservative substitutions (e.g. Leu→Ile, similar chemistry) score positively. Radical substitutions score negatively. This is why protein BLAST is more sensitive than nucleotide BLAST for distant homologues — amino acid chemistry is informative in a way that nucleotide identity alone is not.

In practice: use BLOSUM62 for most searches. The Comparative Genomics course covers matrix selection in depth.

The E-value: The Most Important Number in Your Results

E-value = how many times would you expect to find a match this good (or better) by chance in a database of this size?

`0.0 (≈ 1e-180+)`	Certain	Astronomically unlikely to be coincidental. Essentially certain homologue.
`1e-50`	Very strong	Very strong hit. Highly conserved sequence. Almost certainly true homologue.
`1e-10`	Strong	Strong hit. Likely homologue. Worth investigating.
`1e-5`	Moderate	Moderate. Generally accepted threshold for annotation transfer. Check carefully.
`0.01`	Weak	Weak hit. Treat with scepticism. May be real or may be noise.
`≥ 1.0`	Noise	Expected to occur by chance. Almost certainly not biologically meaningful.

⚠ E-value depends on database size. The same alignment gives a lower E-value against a small database than a large one. Always report which database you searched.

The Five BLAST Programmes

blastn	Query: Nucleotide	DB: Nucleotide	Find similar DNA/RNA sequences. Identify a species from a sequenced fragment (e.g. 16S rRNA, COI barcode). Fast but less sensitive for distant homologues.
blastp	Query: Protein	DB: Protein	Find similar protein sequences. Most sensitive for distant evolutionary relationships. Default choice for functional annotation.
blastx	Query: Nucleotide (translated)	DB: Protein	Find protein matches for an unannotated nucleotide sequence. Translates in all 6 reading frames. Useful when you have a new genomic sequence but no gene model.
tblastn	Query: Protein	DB: Nucleotide (translated)	Find unannotated genomic regions that encode a protein of interest. Useful for identifying new gene models in an unannotated genome.
tblastx	Query: Nucleotide (translated)	DB: Nucleotide (translated)	Most sensitive nucleotide-to-nucleotide search (via protein translation). Slow. Used for finding distant nucleotide homologues where blastn fails.

Interpreting the BLAST Results Table

`Description`

Name of the matching database sequence (subject)

`E value`

Expect value — how likely this match is by chance. Your primary significance indicator.

`Per. ident %`

Percentage of identical residues in the aligned region. Always check alongside query cover.

`Query cover`

Fraction of your query sequence covered by the alignment. 10% coverage at 90% identity ≠ 90% coverage at 90% identity.

`Bit score`

Database-size-independent alignment quality score. Use for comparing across different searches.

`Accession`

Database accession number. Use to fetch the full sequence or annotation.

What You Should Know From This Material

- ✓ Explain what homology means and distinguish orthologues from paralogues
- ✓ Describe the seed-and-extend strategy of BLAST at a conceptual level
- ✓ Explain what an E-value is and interpret one correctly, including the effect of database size
- ✓ Distinguish bit score from E-value and explain when bit score is more appropriate
- ✓ Select the correct BLAST programme for a given query and database type
- ✓ Interpret a BLAST results table: coverage, % identity, E-value, bit score
- ✓ Choose an appropriate substitution matrix (BLOSUM62 default; BLOSUM45/80 for distant/close)
- ✓ Recognise common pitfalls: query coverage vs. percentage identity, and why E-values depend on database size
- ✓ Distinguish orthologues from paralogues in database hits